Finetuning LLMs for CTI (Cyber Threat Intelligence) Domain.

Objective

The primary objective of this project is to fine-tune Large Language Models (LLMs) specifically for the CTI domain. This will involve systematically gathering, preprocessing, and transforming diverse structured and unstructured cybersecurity datasets into formats suitable for model fine-tuning. The goal is to enhance the model's domain-specific capabilities to understand, analyze, and process CTI-related information effectively.

As agreed with you, the project is designed to be executed on a fixed-cost, milestone-based approach, with defined deliverables at each stage.

Milestone 1: Data Collection, Assessment & Usefulness Analysis

Step 1: Data Gathering and Understanding

Data Sources:

Data Type	Count	Formats/Methods
Structured	7	JSON, CSV, STIX
Unstructured	5	RSS Feeds (1), PDF (2), Web Scraping (2)

1. Tools to Scrap the data
2. **Data Gathering:** Collect the data from the identified sources using appropriate methods:
    a. API-based data pulling.
    b. Manual extraction where automation is not feasible.
    c. Writing codes for web scraping wherever applicable.
    d. Understanding the data definition and data dictionary.
    e. Understanding the relationship among various elements within structured data.
3. **Data Usefulness Understanding:** Assess the relevance and applicability of the collected data by analyzing:
    a. Whether the data contains domain-specific information required for model fine-tuning.
    b. If the data can be mapped to relevant cybersecurity concepts such as vulnerabilities, attack techniques, or threat actors.

c. Whether the data needs additional processing, such as translation, normalization, or enrichment.
d. Suitability of the data for transforming into fine-tuning formats, including blog-style narratives, Q&A pairs, or structured knowledge entries.

Deliverables:

1. Report on Quality of Dataset and their usefulness.
2. Report on data cleaning and preprocessing needed.

Milestone 2: Data Cleaning, Preparation & Model Planning

Step 1: Data Cleaning Preprocessing:

a. Clean the collected datasets by removing duplicates, inconsistencies, and irrelevant fields.
b. **LM-based Filtering (if applicable):** Remove irrelevant, noisy, or out-of-domain content using pre-trained language models. For example:
    i. Translate or remove non-english data whichever is applicable.
    ii. Removing Garbage data from the data source such as removing table of content, appendix, and other irrelevant fields.
c. **Heuristic Filtering:** Apply rule-based filtering methods for additional data cleaning.

Step 2: Data Preparation:

For each data source, we will prepare data in a format useful for fine-tuning the model for domain knowledge. For example:

a. Converting the structured data (MITRE, CAPEC etc) into Blog or information style format for model finetuning.
b. Handling Vulnerable data into model finetuning format.
c. Creating some synthetics data in terms of Q&A for task related understanding of the model

Step 3: Model Architecture Strategy Planning

We will plan the initial model design strategy:

- **Model List Identification:** Identify suitable pre-trained models that can be adapted for the project (including pretrained CTI-specific models if applicable).
- **Pretraining vs. Fine-tuning Assessment:** Evaluate the need for additional pretraining versus direct instruction for fine-tuning.
- **Framework Selection:** Identify frameworks and tools to be used for training (with preference for Hugging Face-compatible solutions).
- **Training Platform Identification:** Recommend cloud-based or on-premises platforms suitable for fine-tuning and experimentation (e.g., RunPod).

Note: Data cleaning steps will be performed as needed in the data source and will not be performed where needed.

Deliverables:

1. Cleaned Datasets for Model Finetuning
2. Codebase for Data cleaning and preprocessing
3. List of Models that can be used for finetuning, Platform and framework for finetuning.
4. Infrastructure recommendations (cloud/on-premises) for training.

Milestone 3: Model Fine-Tuning, Infrastructure Setup & Performance Evaluation

1. Codebase & Infrastructure Setup

- Set up the required development environment and codebase for model fine-tuning using Hugging Face libraries and frameworks.
- Configure necessary libraries, dependencies, and utilities for:
    - Parameter-Efficient Fine-Tuning (PEFT) techniques such as **LoRA**, **QLoRA**, or similar methods.
    - Tokenization, data loaders, training scripts, and evaluation pipelines.
- Provision the necessary **cloud-based or on-premises infrastructure** for fine-tuning experiments, such as:
    - **RunPod, Hugging Face Inference Endpoints, or custom cloud instances** (depending on client preference and project needs).

- Document the full environment configuration for reproducibility.

2. Model Fine-Tuning (Domain-Specific Adaptation)

- Fine-tune a **pre-trained LLM** (Selected framework in Milestone 1) to adapt it for **domain-specific knowledge** in CTI.
    - Load the cleaned, structured datasets from Milestone 1.
    - Apply PEFT techniques for efficient fine-tuning.
    - Train and validate the model using the prepared datasets.
    - Tune key hyperparameters (learning rate, batch size, number of epochs, etc.) to optimize performance.

3. Model Evaluation & Benchmarking

- Evaluate the fine-tuned model using **standard open-source benchmarks** relevant to the CTI or cybersecurity domain.
- Assessment criteria may include:
    - Perplexity / Loss metrics.
    - Benchmark Specific metrics.

4. Reporting & Deliverables

- Prepare a comprehensive report summarizing:
    - Fine-tuning methodology and configurations used.
    - Infrastructure setup and environment details.
    - Dataset utilization and processing summary.
    - Model performance metrics and benchmark results.
- Deliverables for this milestone:
    - Fine-tuned model files and code repositories.
    - Configuration and environment documentation.
    - Final evaluation report.

Project Deliverables by Milestone

Milestone	Deliverables	Time Estimate	Considering Parallel Resources
Milestone 1: Data Collection, Assessment & Usefulness Analysis	- Report on Quality of Dataset and their usefulness. - Report on data cleaning and preprocessing needed. - Data Dictionary & Relationship Mapping (for structured data).	58 Days	16 Days
Milestone 2: Data Cleaning, Preparation & Model Planning	- Cleaned Datasets for Model Finetuning - Codebase for Data cleaning and preprocessing - List of Models that can be used for finetuning, Platform and framework for finetuning. - Infrastructure recommendations (cloud/on-premises) for training.	100.5 days	21 Days
Milestone 3: Model Fine-Tuning, Infrastructure Setup & Performance Evaluation	- Fine-tuned model files and code repositories. - Configuration and environment documentation. - Final evaluation report.	18 Days	18 Days
	TOTAL DAYS	**177 Days**	**55 Days**

Final Estimate for Milestone 1:

Task	Time Estimate	Considering Parallel Resource	Cost Estimate (in $)
Data Collection Assessment & Usefulness Analysis	48 Days	12 Days (4 Resource)	9600
Supervision	2 Days	2 Days	480
Buffer Time	8 Days	2 Days (4 Resource)	1600
		TOTAL	11680

Note: Considering the nature of data sources, their availability, and constraints, we have included buffer time of 8 days, which is applicable if additional time is needed.

Final Estimate for Milestone 2:

Task	Time Estimate	Considering Parallel Resource	Cost Estimate (in $)

Data Collection Assessment & Usefulness Analysis	100.5 Days	21 Days (5 Resources)	20100
		TOTAL	20100

Note: Based on the outcome of Milestone 1, the time estimates of the 2nd milestones can change, that will be reevaluated if applicable.

Final Estimate for Milestone 3:

Task	**Time Estimate**	**Considering Parallel Resource**	**Cost Estimate (in $)**
Data Collection Assessment & Usefulness Analysis	15 Days	15 Days (1 Resource)	3600
Buffer Time	3 Days	3 Days (1 Resource)	720
		TOTAL	4320

Note: Based on the outcome of Milestone 1, 2 and size and complexity of the data, the time estimates of the 3rd milestones can change, that will be reevaluated if applicable.

Total Estimate

Milestones	**Cost including buffer(in $)**	**Cost without Buffer (in $)**
Milestone 1	11680	10080
Milestone 2	20100	20100
Milestone 3	4320	3600
Total	36100	33780

Appendices:

Milestone 1: Data Sources are in different formats like GitHub Repositories, Blogs, Links, PDFs. We need to figure out tools for web scraping; we need to write custom codes for web scraping.

Note:

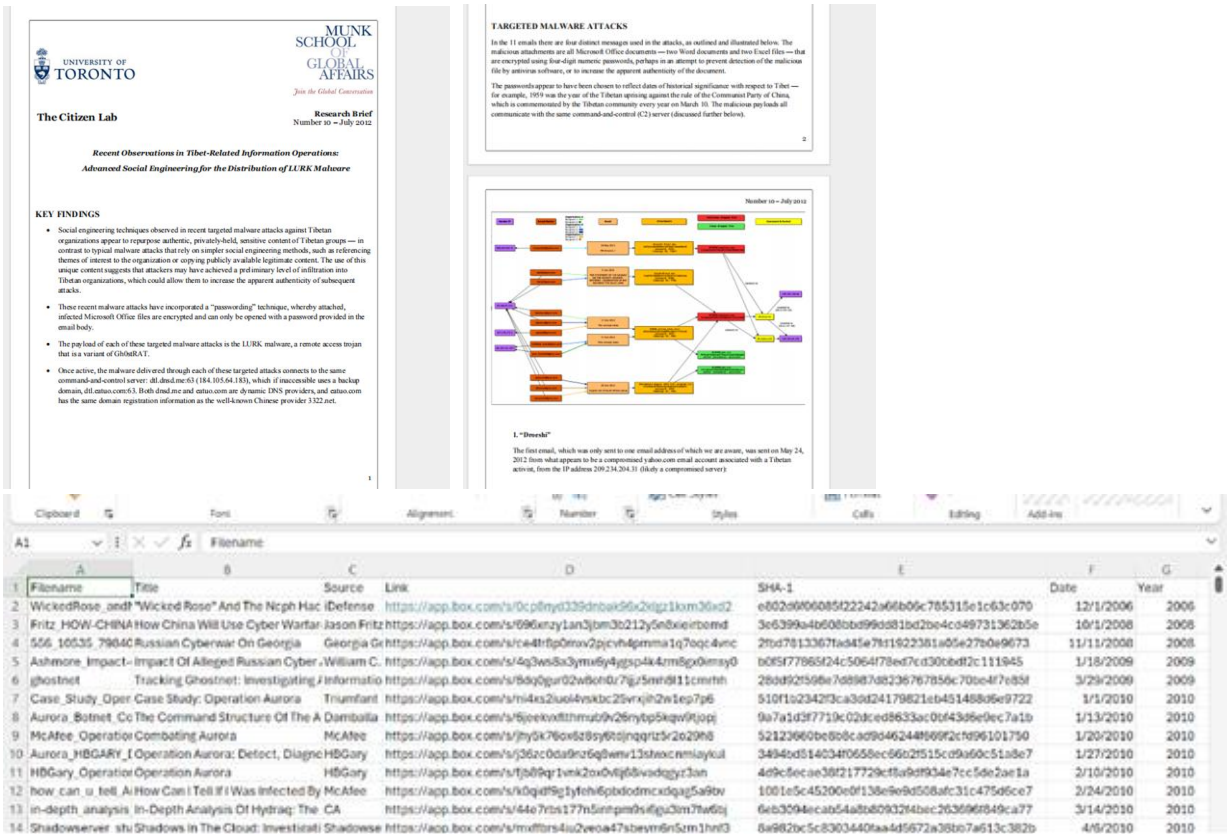
1. Based on the outcome of Milestone 1, the time estimates of the next milestones can change, that will be reevaluated if applicable.

2. Based on our analysis so far, some known complexities in data sources are as follows:

1. **MISP galaxy Data Source:** The MISP Galaxy dataset includes descriptions in different languages, which requires preprocessing. We will need to translate non-English text and filter it to ensure consistency and relevance for further analysis and fine-tuning.

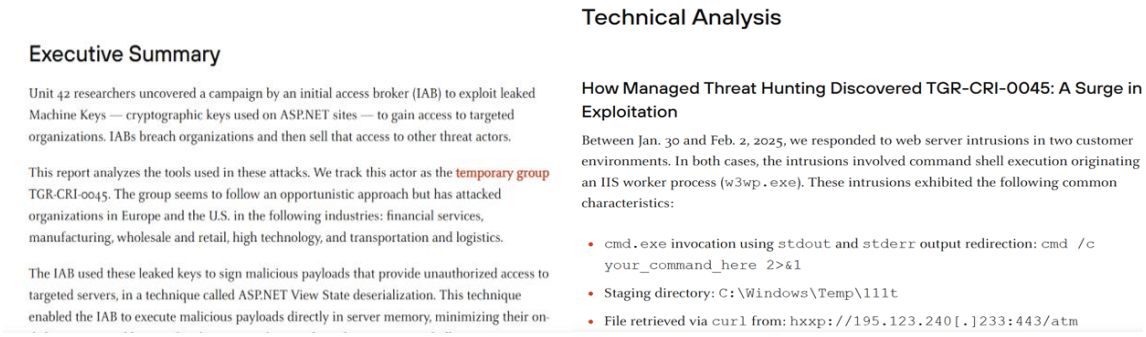

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{
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  "category": "actor",
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          "https://apt.360.net/report/apts/12.html"
        ],
        "suspected-victims": [
          "中国"
        ],
```


2. **APT Notes:** APT Notes are mostly PDF reports containing detailed threat intelligence, including diagrams, charts, and images. Extracting useful information from these PDFs will require careful handling of both text and visual elements to retain full context.



Filename	Title	Source	Link	SHA-1	Date	Year
WickedRose_and	"Wicked Rose" And The Nicph Hac	iDefense	https://app.box.com/s/0cp8nyd339dnbak96x2ktgz1kxm36xd2	e802a9f06085f22242a66b06c765315e1c63c070	12/1/2006	2006
Fritz_HOW-CHINA	How China Will Use Cyber Warfar	Jason Fritz	https://app.box.com/s/696xnzy1an3jbm3b212y5n8xieirbomd	3e6399a4b608bbd99dd81bd2be4cd49731362b5e	10/1/2008	2008
556_16535_79840	Russian Cyberwar On Georgia	Georgia Gi	https://app.box.com/s/ce4frfip0mxv2pjcvh4pmma1q7oqc4vnc	2fbd7813367fad45e7fd1922381a05e27b0e9673	11/11/2008	2008
Ashmore_Impact-	Impact Of Alleged Russian Cyber	William C.	https://app.box.com/s/4q3ws8x3ymu6y4pgsp4k4rmi8gx0kmsy0	b0f5f77865f24c5064f78ed7cd30cbdf2c111945	1/18/2009	2009
ghostnet	Tracking Ghostnet: Investigating A	Informatio	https://app.box.com/s/8dq0gur02w8oh0z7ijz5mh9l11cmrhh	28dd92f598e7d898f7d8236767858c70be4f7e85f	3/29/2009	2009
Case_Study_Oper	Case Study: Operation Aurora	Triumfant	https://app.box.com/s/ni4xs2iuoi4vskbc25vrxjih2w1ep7p6	510f1b2342f3ca3dd24179821eb451488d6e9722	1/1/2010	2010
Aurora_Botnet_Cc	The Command Structure Of The A	Damballa	https://app.box.com/s/6jeekvxfithmutb9v26nybp5kqw9tjopj	9a7a1d3f7719c02dced8633ac0bf43d6e0ec7a1b	1/13/2010	2010
McAfee_Operatior	Combating Aurora	McAfee	https://app.box.com/s/jhy0k76ox6z8sy6tdjnqqriz5r2o29h8	52123660be8b8cad9d46244f669f2cfd96101750	1/20/2010	2010
Aurora_HBGARY_[	Operation Aurora: Detect, Diagnc	HBGary	https://app.box.com/s/j36zc0da9nz6q8wnv13shxxcnmiaykul	3494bd814034f0658ec66b2f515cd9a66c51a8e7	1/27/2010	2010
HBGary_Operatior	Operation Aurora	HBGary	https://app.box.com/s/tjb09qr1vnk2ox0vtjd8vadqgyz3an	4d9c8ecae38f217729cffa9df934e7cc5de2ae1a	2/10/2010	2010
how_can_u_tell_Ai	How Can I Tell If I Was Infected By	McAfee	https://app.box.com/s/k0qidf9g1yfehi6pbdiodmcxdqag5a9bv	1001e5c45200e0f138e9e9d508afc31c475d6ce7	2/24/2010	2010
in-depth_analysis	In-Depth Analysis Of Hydraq: The	CA	https://app.box.com/s/44e7rbs177n5intpm9s6gu3tm7fw6bj	6eb3094ecab54a8b8093264bec263696f849ca77	3/14/2010	2010
Shadowserver_shi	Shadows In The Cloud: Investigati	Shadowse	https://app.box.com/s/mxffbrs4iu2veoa47sbevm6n5zm1hnf3	8a982bc5c8303440faa4d5672a38bb7a613c382b	4/6/2010	2010

3. **CTI Blogs/ Exploit DB:** Some blogs and exploit databases contain important command-line data embedded as images instead of text. Extracting this information accurately will require web scraping along with OCR techniques to ensure that critical technical details are not lost.



Executive Summary

Unit 42 researchers uncovered a campaign by an initial access broker (IAB) to exploit leaked Machine Keys — cryptographic keys used on ASP.NET sites — to gain access to targeted organizations. IABs breach organizations and then sell that access to other threat actors.

This report analyzes the tools used in these attacks. We track this actor as the temporary group TGR-CRI-0045. The group seems to follow an opportunistic approach but has attacked organizations in Europe and the U.S. in the following industries: financial services, manufacturing, wholesale and retail, high technology, and transportation and logistics.

The IAB used these leaked keys to sign malicious payloads that provide unauthorized access to targeted servers, in a technique called ASP.NET View State deserialization. This technique enabled the IAB to execute malicious payloads directly in server memory, minimizing their on-

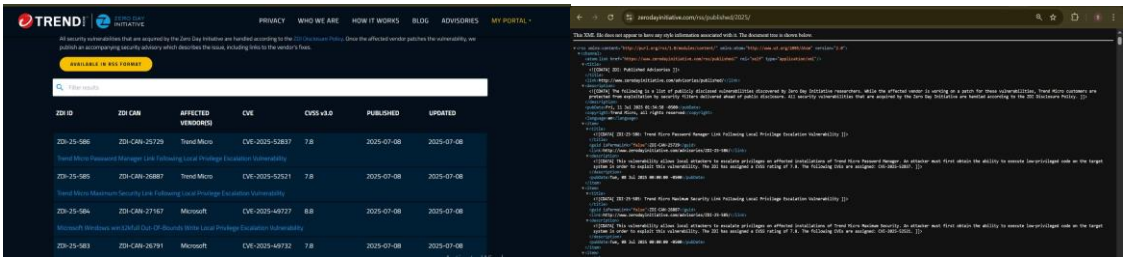
Technical Analysis

How Managed Threat Hunting Discovered TGR-CRI-0045: A Surge in Exploitation

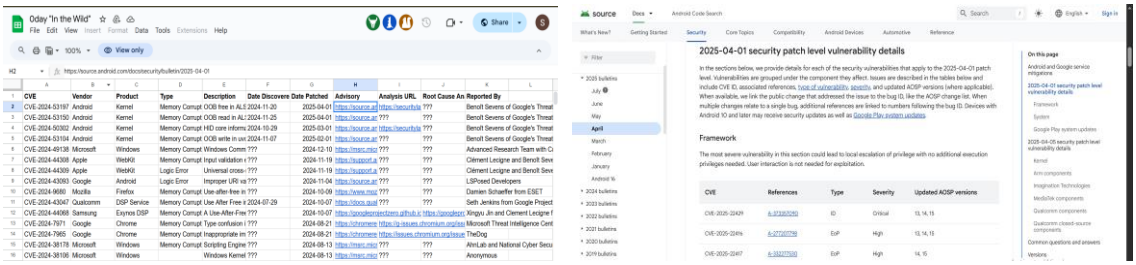
Between Jan. 30 and Feb. 2, 2025, we responded to web server intrusions in two customer environments. In both cases, the intrusions involved command shell execution originating an IIS worker process (`w3wp.exe`). These intrusions exhibited the following common characteristics:

- `cmd.exe` invocation using `stdout` and `stderr` output redirection: `cmd /c your_command_here 2>&1`
- Staging directory: `C:\Windows\Temp\111t`
- File retrieved via `curl` from: `hxxp://195.123.240[.]233:443/atm`

4. Zero Day Initiatives: ZDI advisories are provided in RSS format, which requires parsing and conversion into a structured format. Additional steps will be needed to fetch detailed advisory content from external links for complete data extraction.



5. Google Project Zero: Vulnerability advisories from Google Project Zero are hosted as unstructured web pages, requiring targeted web scraping and data parsing to collect and organize relevant cybersecurity information.



3. Efforts Time calculation

Data Source	Download / Web Scraping	Usefulness Data Understanding	Data Dictionary Understanding	Relationship Understanding	Data Preprocessing Understanding	Documentation	Total Days required
MITRE ATT&CK	0.5	1	0.5	1	1	0.5	**4.5**
CAPEC	1.5	0.5	0	0.5	1	0.5	**4**
MISP-Galaxy	0.5	0	1	1	1	0.5	**4**
MISP-Texonomies	0.5	0.5	1	0	1	0.5	**3.5**

GOOGLE PROJECT ZERO	2	0	1	0	1	0.5	**4.5**
CWE	0	1	0	1	1	0.5	**3.5**
CVE	0	0	1	1	1	0.5	**3.5**
ZDI	0.5	1	0	1	1	0.5	**4**
THREATFOX IOC	1.5	0	1	0.5	1	0.5	**4.5**
APT NOTES	2	0.5	0	0	1	0.5	**4**
CTI BLOGS	2	0	0.5	0	1	0.5	**4**
Exploit DB	2	0	0.5	0	1	0.5	**4**
Buffer Time							**10**
					TOTAL DAYS		**58**

4. Assumptions & Uncertainties that might occur:

- **Data Accessibility:** Assumes all data sources remain accessible; delays may occur if approvals or access issues arise.
- **Format Consistency:** Estimates are based on current formats; unexpected variations may increase effort.
- **Manual Work:** Web scraping estimates assume minimal restrictions; extra effort may be needed for complex sites.
- **Multilingual Data:** Time for translation is estimated; more effort may be required if more non-English data is found.
- **Documentation Availability:** Assumes basic documentation exists; missing or unclear documentation may need extra analysis.
- **Data Overlaps:** Some sources may overlap, requiring additional deduplication.
- **Scope Stability:** Estimates assume no additional datasets or scope changes during this milestone.

MILESTONE 2:

1. Based on the outcome of Milestone 1, the time estimates of the next milestones can change, that will be reevaluated if applicable.

2. We will have to assess the complexity after the outcome of Milestone 1 for data processing and cleaning.

3. Efforts Time calculation

Tasks	Filtering	Deduplication	Garbage Removing	Heuristic Filtering	Dataset Preparation	Intra Corelationship	Q&A Data Preparation	Days Required
MITRE ATT&CK		1	1	1	2.5	1	2	8.5
CAPEC	1	1	1	1	2.5	1	1	7.5
MISP-Galaxy	1	1	1	1	2.5	1	1	7.5
MISP-Texonomies	1	1	1	1	2.5	1	1	7.5
GOOGLE PROJECT ZERO	1	1	1	1	2.5	1	2	8.5
CWE	1	1	1	1	2.5	1	1	7.5
CVE	1	1	1	1	2.5	1	2	8.5
ZDI	1	1	1	1	2.5	1	1	7.5
THREATFOX IOC	1	1	1	1	2.5	1	1	7.5
APT NOTES	1	1	1	1	2	1	1	7
CTI BLOGS	1	1	1	1	2	1	1	7
Exploit DB	1	1	1	1	2	1	2	8
Model Selection								2
Pretraining vs. Fine-tuning Assessment								2
Framework Selection								2
Training Platform Identification								2
					TOTAL DAYS			**100.5**

MILESTONE 3:

1. Based on the outcome of Milestone 1, the time estimates of the 3rd milestone can change, that will be reevaluated if applicable.

2. We are considering that evaluation datasets will be provided by you.

3. And Based on the tasks in the evaluation the time can vary.